

Module-3

Combinatorics

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Counting in Probability

- For a finite sample space S with equally likely outcomes, the probability of an event A is given by:

$$P(A) = \frac{|A|}{|S|} = \frac{M}{N}$$

- Finding the probability of A involves counting the elements in A and S .
- We will see efficient methods for counting elements in a set. Specifically:
 - Introduction to basic counting concepts relevant to probability.
 - The multiplication principle which is key to counting.
 - Revisit the multiplication principle from a new perspective.



Example-1

Suppose that I want to purchase a tablet computer. I can choose either a large or a small screen; a 64GB, 128 GB, or 256 GB storage capacity, and a black or white cover. How many different options do I have?



Example-1 (Solution)

$$\text{Answer} = 3 \times 3 \times 2 = 12$$

① L-64-B

② L-64-W

③ L-128-B

④ L-128-W

⑤ L-256-B

⑥ L-256-W

⑦ S-64-B

⑧ S-64-W

⑨ S-128-B

⑩ S-128-W

⑪ S-256-B

⑫ S-256-W



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Counting Terminology in Probability

- **Sampling:** Choosing an element from a set, often at random with equal probability for each element.
- **With or Without Replacement:**
 - *With replacement:* Elements can be chosen multiple times; repetition is allowed.
 - *Without replacement:* Elements can be chosen only once; repetition is not allowed.
- **Ordered vs. Unordered Sampling:**
 - *Ordered:* The order of selection matters (e.g., $a_1, a_2, a_3 \neq a_2, a_3, a_1$).
 - *Unordered:* The order of selection does not matter.



Counting Terminology in Probability (Cont'd)

- **Four Sampling Types:**

- ① **Ordered sampling with replacement**
 - ② **Ordered sampling without replacement**
 - ③ **Unordered sampling with replacement**
 - ④ **Unordered sampling without replacement**
- Formulas for each type will be provided, with a focus on understanding their derivation rather than memorization.



Ordered Sampling with Replacement



Ordered Sampling with Replacement

- Set contains n elements (e.g., $A = \{1, 2, 3, \dots, n\}$).
- Goal: Draw k samples such that:
 - Order matters.
 - Repetition is allowed.
- Example: For $A = \{1, 2, 3\}$ and $k = 2$, there are 9 possibilities:
 - $(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)$
- General rule:
 - There are k positions to fill.
 - Each position has n possible options.
- Total number of ways to choose k objects from a set of n elements:

$$n \times n \times n \dots \times n = n^k$$

- This is a special case of the multiplication principle: k "experiments" with n outcomes each.



Example-2

I need to choose a password for a computer account. The rule is that the password must consist of two lowercase letters (a to z) followed by one capital letter (A to Z) followed by four digits (0, 1, . . . , 9). For example, the following is a valid password:

ejT3018

Find the total number of possible passwords, N .

A hacker has been able to write a program that randomly and independently generates 10^8 passwords according to the above rule. Note that the same password could be generated more than once. If one of the randomly chosen passwords matches my password, then he can access my account information. What is the probability that he is successful in accessing my account information?



Example-2 (Solution)

passwords: $N = 26 \times 26 \times 26 \times 10^4 = 26^3 \times 10^4$

G_i : hacker's i -th guess matches my password, where $i = 1, 2, \dots, 10^8$.

$$P(G_i) = \frac{1}{N}$$

Let p_{hack} be the probability that the hacker is successful (at least one guess matches my password). Since the guesses are independent:

$$p_{\text{hack}} = P\left(\bigcup_i G_i\right)$$

It is easier to calculate the complement (none of the guesses match) and then subtract from 1:

$$P\left(\bigcup_i G_i\right)^c = P\left(\bigcap_i G_i^c\right) = \prod_{i=1}^{10^8} P(G_i^c) = \left(1 - \frac{1}{N}\right)^{10^8}$$

Thus,

$$p_{\text{hack}} = 1 - \left(1 - \frac{1}{N}\right)^{10^8} = 1 - \left(1 - \frac{1}{26^3 \times 10^4}\right)^{10^8} \approx 0.4339$$



Ordered Sampling Without Replacement



Ordered Sampling without Replacement

- Set contains n elements (e.g., $A = \{1, 2, 3, \dots, n\}$).
- Goal: Draw k samples such that:
 - Order matters.
 - Repetition is not allowed.
- Example: For $A = \{1, 2, 3\}$ and $k = 2$, there are 6 possibilities:
 - $(1, 2), (1, 3), (2, 1), (2, 3), (3, 1), (3, 2)$
- General rule:
 - k positions to fill.
 - n options for the first position, $(n - 1)$ for the second, $(n - 2)$ for the third, and so on.
 - Total number of ways to choose k objects without replacement:

$$n \times (n - 1) \times \cdots \times (n - k + 1) = \frac{n!}{(n - k)!} = P_{n,k}$$



Birthday Problem

Example: If k people are at a party, what is the probability that at least two of them have the same birthday? Assume there are $n = 365$ days in a year.



Birthday Problem (Cont'd)

Example: If k people are at a party, what is the probability that at least two of them have the same birthday? Assume there are $n = 365$ days in a year.

Solution: Let A be the event that at least two people share the same birthday. We will first find the complement, $P(A^c)$, where no two people have the same birthday:

$$P(A) = 1 - P(A^c).$$

The total number of possible birthday sequences is n^k . If no two birthdays are the same, the number of ways to assign birthdays is

$P(n, k) = n \times (n - 1) \times \cdots \times (n - k + 1)$. Thus, the probability is:

$$P(A) = 1 - \frac{P(n, k)}{n^k}.$$



Birthday Problem (Cont'd)

- This is known as the birthday paradox because the probability is often higher than expected. For $k = 23$, the probability that at least two people have the same birthday is $P(A) = 0.5073$. The probability crosses 99% when there are 57 people.
- People often think of this problem in terms of comparing their own birthday with others (which has a low chance).
- However, the problem considers all possible pairs of people, not just one person comparing their birthday to others.
- As the number of people increases, the number of comparisons grows rapidly, leading to a much higher probability than expected.



Unordered Sampling without Replacement



Unordered Sampling without Replacement: Combinations

- Set contains n elements (e.g., $A = \{1, 2, 3, \dots, n\}$).
- Goal: Draw k samples such that:
 - Order does not matter.
 - Repetition is not allowed.
- We seek a k -element subset of A , known as a k -combination.
- Example: For $A = \{1, 2, 3\}$ and $k = 2$, there are 3 possibilities:
 - $\{1, 2\}, \{1, 3\}, \{2, 3\}$
- The number of k -element subsets of A is denoted by $\binom{n}{k}$.
- Relation to k -permutations $P_{n,k}$:

$$P_{n,k} = \binom{n}{k} \times k!$$



Unordered Sampling without Replacement: Combinations

- Formula for combinations:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

- $\binom{n}{k}$ is also called the binomial coefficient.
- Binomial theorem: For $n \geq 0$,

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k}$$

- Note: Various notations for combinations include $C_{n,k}$, $C(n, k)$, C_k^n , nCk , etc. This text uses $\binom{n}{k}$.



Example: Probability with a Deck of Cards

- **Problem:** Choose 3 cards from a standard deck of 52 cards. Find the probability that these cards contain at least one ace.
- **Solution:** ??



Example: Probability with a Deck of Cards (Solution)

- Let A be the event that at least one ace is chosen, and A^c be the event that no aces are chosen.
- Sample space $|S|$: All ways to choose 3 cards from 52: $|S| = \binom{52}{3}$
- Number of outcomes in A^c (choosing 3 non-ace cards):
 - There are 48 non-ace cards in the deck (52 total - 4 aces).
 - Ways to choose 3 non-ace cards: $|A^c| = \binom{48}{3}$
- Probability of A^c (no aces chosen): $P(A^c) = \frac{|A^c|}{|S|} = \frac{\binom{48}{3}}{\binom{52}{3}}$
- Probability of A (at least one ace):

$$P(A) = 1 - P(A^c) = 1 - \frac{\binom{48}{3}}{\binom{52}{3}}$$



Example: Distinct Sequences with Letters

- **Problem:** How many distinct sequences can be made using 3 letter "A"s and 5 letter "B"s? (Examples: AAABBBBB, AABABBBB, etc.)
- **Solution:** ??



Example: Distinct Sequences with Letters (Cont'd)

- **Problem:** How many distinct sequences can be made using 3 letter "A"s and 5 letter "B"s? (Examples: AAABBBBB, AABABBBB, etc.)

- **Solution:**

- Total positions to fill with letters A or B: $3 + 5 = 8$.
- Choose 3 positions for "A"s out of 8:

$$\binom{8}{3}$$

- Remaining positions will be filled with "B"s.
- Alternatively, choose 5 positions for "B"s out of 8:

$$\binom{8}{5}$$

- Since choosing positions for "A"s or "B"s is equivalent:

$$\binom{8}{3} = \binom{8}{5}$$



Example: Combinatorial Identities

- **Problem:** Show the following identities for non-negative integers k , m , and n using combinatorial arguments:

① $\sum_{k=0}^n \binom{n}{k} = 2^n.$

② For $0 \leq k < n$, $\binom{n+1}{k+1} = \binom{n}{k+1} + \binom{n}{k}.$

③ Vandermonde's identity: $\binom{m+n}{k} = \sum_{i=0}^k \binom{m}{i} \binom{n}{k-i}.$



Example: Combinatorial Identities (Cont'd)

- To prove $\sum_{k=0}^n \binom{n}{k} = 2^n$:
 - Count total subsets of an n -element set A : 2^n .
 - Alternatively, sum subsets of each size k : $\sum_{k=0}^n \binom{n}{k}$.
 - Using binomial theorem: $(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k}$.
 - Set $a = b = 1$ to get $2^n = \sum_{k=0}^n \binom{n}{k}$.
- To prove $\binom{n+1}{k+1} = \binom{n}{k+1} + \binom{n}{k}$:
 - Consider an arbitrary set A with $n + 1$ elements:
 $A = \{a_1, a_2, \dots, a_{n+1}\}$.
 - Choose $k + 1$ -element subset B in $\binom{n+1}{k+1}$ ways.
 - Divide into cases:
 - Case 1: $a_{n+1} \notin B$, choose $k + 1$ elements from $\{a_1, \dots, a_n\}$: $\binom{n}{k+1}$ ways.
 - Case 2: $a_{n+1} \in B$, choose k elements from $\{a_1, \dots, a_n\}$: $\binom{n}{k}$ ways.
 - Or just add these two: $\binom{n}{k+1} + \binom{n}{k}$.



Example: Combinatorial Identities (Cont'd)

- To prove Vandermonde's identity $\binom{m+n}{k} = \sum_{i=0}^k \binom{m}{i} \binom{n}{k-i}$:
 - Consider a set A with $m+n$ elements:
 $A = \{a_1, a_2, \dots, a_m, b_1, b_2, \dots, b_n\}$.
 - Count k -element subsets of A : $\binom{m+n}{k}$.
 - Alternatively, choose i elements from $\{a_1, \dots, a_m\}$ and $k-i$ elements from $\{b_1, \dots, b_n\}$.
 - Total: $\sum_{i=0}^k \binom{m}{i} \binom{n}{k-i}$.



Another Interpretation of $\binom{n}{k}$

- Suppose we have a group of n people. We want to divide them into two groups, A and B , where:
 - Group A consists of k people.
 - Group B consists of $n - k$ people.
- The number of ways to choose k people for group A (and hence the rest for group B) is given by $\binom{n}{k}$.
- Therefore, $\binom{n}{k}$ represents the number of ways to divide n distinct objects into two groups of size k and $n - k$.



Another Interpretation of $\binom{n}{k}$ (Cont'd)

- **Special Case:** When $n = 2k$ and we do not care about the names of the groups:
 - The number of ways to divide the objects is $\frac{1}{2} \binom{n}{k}$ to avoid double counting.
 - Example: Dividing 22 players into two soccer teams of 11 players each results in $\frac{1}{2} \binom{22}{11}$ ways.
 - This is because the choices $A = \{1, 2, 3, \dots, 11\}$ and $B = \{12, 13, 14, \dots, 22\}$ are equivalent to $A = \{12, 13, 14, \dots, 22\}$ and $B = \{1, 2, 3, \dots, 11\}$.



Alternate Way To Solve Previous Problem

- **Problem:** How many distinct sequences can be made using 3 letter "A"s and 5 letter "B"s? (Examples: AAABBBBB, AABABBBB, etc.)
 - We have 8 blank positions to be filled with letters "A" or "B."
 - Dividing the positions into two groups: A with 3 positions and B with 5 positions.
 - Just fill the "A"s in $\binom{8}{3}$ ways the remaining will be occupied by "B"
 - The total number of possibilities is $\binom{8}{3}$.



Bernoulli Trials and Binomial Distribution

• Bernoulli Trials:

- A Bernoulli trial is a random experiment with exactly two possible outcomes.
- Outcomes are typically labeled as "success" and "failure."
- Examples:
 - Tossing a coin: "heads" (success) or "tails" (failure).
 - Taking a pass-fail test: "pass" (success) or "fail" (failure).
- The probability of success is denoted by p , and the probability of failure is $q = 1 - p$.

• Binomial Distribution:

- A binomial experiment consists of n independent Bernoulli trials.
- We are interested in the total number of successes in these n trials.
- Example:
 - Tossing a coin n times and counting the number of heads.



Example: Coin Tosses and Probability

- **Given:** A coin with $P(H) = p$ and $P(T) = 1 - p$. Toss the coin 5 times.



Probability of Specific Outcomes

- **THHHH:**

$$P(\text{THHHH}) = p(T) \times p(H) \times p(H) \times p(H) \times p(H) = (1 - p) \cdot p^4$$

- **HTHHH:**

$$\begin{aligned} P(\text{HTHHH}) &= p(H) \times p(T) \times p(H) \times p(H) \times p(H) = \\ &p \cdot (1 - p) \cdot p^3 = (1 - p) \cdot p^4 \end{aligned}$$

- **HHTHH:**

$$\begin{aligned} P(\text{HHTHH}) &= p(H) \times p(H) \times p(T) \times p(H) \times p(H) = \\ &p^2 \cdot (1 - p) \cdot p^3 = (1 - p) \cdot p^4 \end{aligned}$$



Exactly Four Heads and One Tail

- Let B be the event of observing exactly one tail and four heads:

$$B = \{T H H H H, H T H H H, H H T H H, H H H T H, H H H H T\}$$

$$P(B) = P(T H H H H) + P(H T H H H) + P(H H T H H) + P(H H H T H) + P(H H H H T)$$

$$P(B) = 5 \cdot (1 - p) \cdot p^4$$



Exactly Three Heads and Two Tails

- Let C be the event of observing exactly three heads and two tails:

$$C = \{TTHHH, THTHH, THHTH, \dots, HHHTT\}$$

$$P(C) = |C| \cdot p^3 \cdot (1 - p)^2$$

- The number of distinct sequences with three heads and two tails:

$$|C| = \binom{5}{3}$$

$$P(C) = \binom{5}{3} \cdot p^3 \cdot (1 - p)^2$$



- For n tosses, the probability of observing exactly k heads and $n - k$ tails:

$$P(k \text{ heads}) = \binom{n}{k} \cdot p^k \cdot (1 - p)^{n-k}$$



Binomial Formula

- For n independent Bernoulli trials, where each trial has a success probability p , the probability of observing exactly k successes is given by:

$$P(k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

- Where:**

- $\binom{n}{k}$ is the number of ways to choose k successes out of n trials.
- p^k is the probability of k successes.
- $(1 - p)^{n-k}$ is the probability of $n - k$ failures.



Example: Binomial Formula

- Suppose we flip a coin 5 times, with p being the probability of heads.
- To find the probability of exactly 3 heads and 2 tails:

$$P(3 \text{ heads}) = \binom{5}{3} p^3 (1 - p)^2$$

- Here:

$$\binom{5}{3} = \frac{5!}{3!(5-3)!} = 10$$

$$P(3 \text{ heads}) = 10 \cdot p^3 \cdot (1 - p)^2$$



Generalization to Multinomial Coefficients

- The binomial coefficient $\binom{n}{k}$ represents the number of ways to divide n objects into two groups: one of size k and one of size $n - k$.
- This concept generalizes to dividing n objects into more than two groups.
- For n objects divided into m groups with sizes $k_1, k_2, \dots, k_m$, where $\sum_{i=1}^m k_i = n$:

$$\text{Number of ways} = \frac{n!}{k_1! \cdot k_2! \cdot \dots \cdot k_m!}$$

- This is known as the multinomial coefficient.



Example: Multinomial Coefficients

- Suppose we have 10 objects and we want to divide them into 3 groups of sizes 2, 3, and 5.
- The number of ways to do this is:

$$\frac{10!}{2! \cdot 3! \cdot 5!}$$

- This represents the number of ways to distribute 10 distinct objects into three distinct groups with 2, 3, and 5 objects, respectively.



Multinomial Theorem

- The multinomial theorem generalizes the binomial theorem to more than two terms.
- It states:

$$(x_1 + x_2 + \cdots + x_r)^n = \sum_{n_1+n_2+\cdots+n_r=n} \binom{n}{n_1, n_2, \dots, n_r} x_1^{n_1} x_2^{n_2} \cdots x_r^{n_r}$$

- Here, $\binom{n}{n_1, n_2, \dots, n_r}$ represents the multinomial coefficient.
- For Example: $(x_1 + x_2 + x_3)^4$, the expansion is:

$$(x_1 + x_2 + x_3)^4 = \sum_{n_1+n_2+n_3=4} \binom{4}{n_1, n_2, n_3} x_1^{n_1} x_2^{n_2} x_3^{n_3}$$

- Expanding this, we get terms like x_1^4 , $x_1^3 x_2$, $x_1^2 x_2^2$, etc.



Extended Binomial Formula

- The binomial formula for Bernoulli trials can be extended to cases with more than two possible outcomes.
- For an experiment with n trials and r possible outcomes, the probability of observing k_1 outcomes of type 1, k_2 outcomes of type 2, ..., k_r outcomes of type r (with $\sum_{i=1}^r k_i = n$) is given by:

$$P(k_1, k_2, \dots, k_r) = \frac{n!}{k_1! \cdot k_2! \cdot \dots \cdot k_r!} p_1^{k_1} p_2^{k_2} \dots p_r^{k_r}$$

- Here, p_i represents the probability of the i -th outcome.



Example:

I roll a die 18 times. What is the probability that each number appears exactly 3 times?



Step 1: Probability for One Sequence

First, let's calculate the probability of a specific sequence of outcomes where each number appears exactly 3 times.

Since a die has 6 faces, the probability of rolling any number on a single roll is $\frac{1}{6}$.

The probability of a sequence where each number appears 3 times (e.g., three 1's, three 2's, and so on) is:

$$\left(\frac{1}{6}\right)^3 \times \left(\frac{1}{6}\right)^3 \times \left(\frac{1}{6}\right)^3 \times \left(\frac{1}{6}\right)^3 \times \left(\frac{1}{6}\right)^3 \times \left(\frac{1}{6}\right)^3 = \left(\frac{1}{6}\right)^{18}.$$



Step 2: Number of Possible Sequences

Now, let's calculate how many distinct sequences are possible where each number appears exactly 3 times.

We have 18 positions, and we need to assign exactly 3 positions for each of the numbers 1 through 6. This is a classic multinomial problem.

The number of ways to assign the positions is given by the multinomial coefficient:

$$\binom{18}{3, 3, 3, 3, 3, 3} = \frac{18!}{3! \times 3! \times 3! \times 3! \times 3! \times 3!}.$$



Step 3: Total Probability

Now, to find the total probability, we multiply the probability of one specific sequence by the number of distinct sequences.

The total probability is:

$$P = \frac{18!}{(3!)^6} \times \left(\frac{1}{6}\right)^{18}.$$

This gives us the exact probability that each number on the die appears exactly 3 times when the die is rolled 18 times.



Unordered Sampling with Replacement



Unordered Sampling with Replacement

Problem: We want to sample from the set $A = \{a_1, a_2, \dots, a_n\}$ k times, such that repetition is allowed and ordering does not matter.

Example: If $A = \{1, 2, 3\}$ and $k = 2$, there are 6 different ways to do this:

1, 1; 1, 2; 1, 3; 2, 2; 2, 3; 3, 3.



Representation

Any pair can be represented by the number of 1's, 2's, and 3's it contains. Let x_1 be the number of ones, x_2 the number of twos, and x_3 the number of threes. Each pair can be represented by a vector (x_1, x_2, x_3) , e.g.:

$$1, 1 \rightarrow (x_1, x_2, x_3) = (2, 0, 0)$$

$$1, 2 \rightarrow (x_1, x_2, x_3) = (1, 1, 0)$$

$$1, 3 \rightarrow (x_1, x_2, x_3) = (1, 0, 1)$$



Generalization

For any set $A = \{1, 2, \dots, n\}$ and sample size k , the number of unordered k -samples with replacement is the number of solutions to:

$$x_1 + x_2 + \dots + x_n = k, \quad \text{where } x_i \geq 0.$$

Theorem: The number of distinct solutions is given by:

$$\binom{n+k-1}{k} = \binom{n+k-1}{n-1}.$$



Proof (Sketch)

Each x_i can be represented by vertical lines:

$$1 \rightarrow |, \quad 2 \rightarrow ||, \quad 3 \rightarrow |||, \dots$$

For any solution to $x_1 + x_2 + \dots + x_n = k$, we use k vertical lines and $n - 1$ plus signs. This gives:

$$\binom{n + k - 1}{k}.$$



Example: Shuttle Passengers and Hotels

Problem: Ten passengers board an airport shuttle that serves 5 hotels. The shuttle driver records how many passengers get off at each hotel. How many different possibilities exist for the distribution of passengers across the hotels?



Solution: Defining Variables

Let x_i represent the number of passengers getting off at Hotel i where $i = 1, 2, 3, 4, 5$. Thus, we have the equation:

$$x_1 + x_2 + x_3 + x_4 + x_5 = 10, \quad \text{where } x_i \geq 0.$$

The number of distinct possibilities corresponds to the number of solutions to the equation:

$$x_1 + x_2 + x_3 + x_4 + x_5 = 10,$$

which is a classic example of *unordered sampling with replacement*.



Solution: Formula

Using the formula for the number of solutions to the equation:

$$x_1 + x_2 + \cdots + x_n = k,$$

the number of distinct solutions is given by:

$$\binom{n + k - 1}{k} = \binom{5 + 10 - 1}{10} = \binom{14}{4}.$$



Solution: Formula

We have a set with n elements, and we want to draw k samples from the set:

Sampling Method	Formula
Ordered sampling with replacement	n^k
Ordered sampling without replacement	$P_{n,k} = \frac{n!}{(n-k)!}$
Unordered sampling without replacement	$\binom{n}{k} = \frac{n!}{k!(n-k)!}$
Unordered sampling with replacement	$\binom{n+k-1}{k}$



Do It Yourself

Problem-1: Let A and B be two finite sets, with $|A| = m$ and $|B| = n$. How many distinct functions (mappings) can you define from set A to set B , $f : A \rightarrow B$?

Problem-2: A function is said to be *one-to-one* if for all $x_1 \neq x_2$, we have $f(x_1) \neq f(x_2)$. Equivalently, we can say a function is one-to-one if whenever $f(x_1) = f(x_2)$, then $x_1 = x_2$. Let A and B be two finite sets, with $|A| = m$ and $|B| = n$. How many distinct one-to-one functions (mappings) can you define from set A to set B , $f : A \rightarrow B$?

Problem-3: An urn contains 30 red balls and 70 green balls. What is the probability of getting exactly k red balls in a sample of size 20 if the sampling is done with replacement (repetition allowed)? Assume $0 \leq k \leq 20$.



Thanks

